

Unit-3 Introduction to graph theory: Königsberg's bridge problems, Utilities problem, Seating Problem, Graphs, Representation of Graphs. Directed graphs, Incidence, Adjacency, Degree, in-degree, out-degree, Regular graphs, Complete graphs, Null Graph, Bipartite Graphs, Isomorphism, Directed Graphs, Sub graphs, Walk, Trail, Path, Circuit, Cycle, Connected and disconnected graphs.

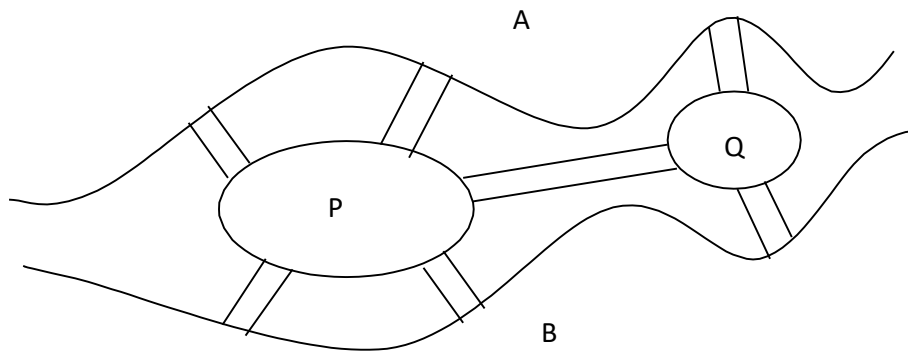
The Königsberg Bridge Problem:

In the 19th century, a city named Königsberg in East Prussia (Europe) was situated on the Pregel River, which divided the city into four parts. The two of these parts were the banks of the river and two were islands. These parts were connected with each other through SEVEN bridges as depicted in the figure. The citizens of the city seemed to have the following problem by starting at any of the four land area, can we return to that area after crossing each of the seven bridges exactly once?

The problem, known as the Königsberg bridge problem and remained unsolved for several years. In 1936 Euler analysed the problem with the help of a graph and gave the solution. This was indeed the starting point for the development of the graph theory.

(Discuss the Königsberg bridge problem and the solution of the problem)

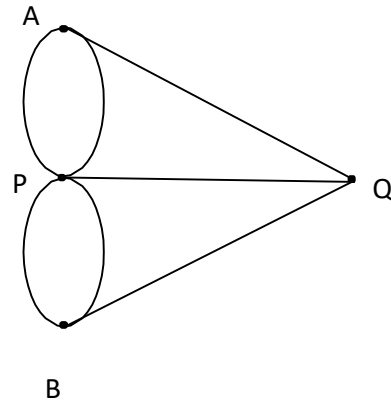
There are two islands and two land areas with seven bridges between them, as shown in the figure



The problem is to start at any four areas of the city, A, B, P, Q walk over any of the seven bridges exactly once and return to the starting point.

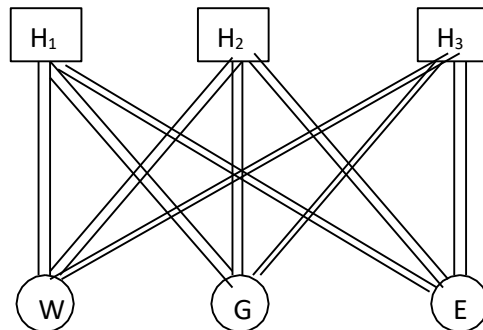
Solution:

Construct a graph by denoting the four land areas as four vertices and the seven bridges connecting them as seven edges



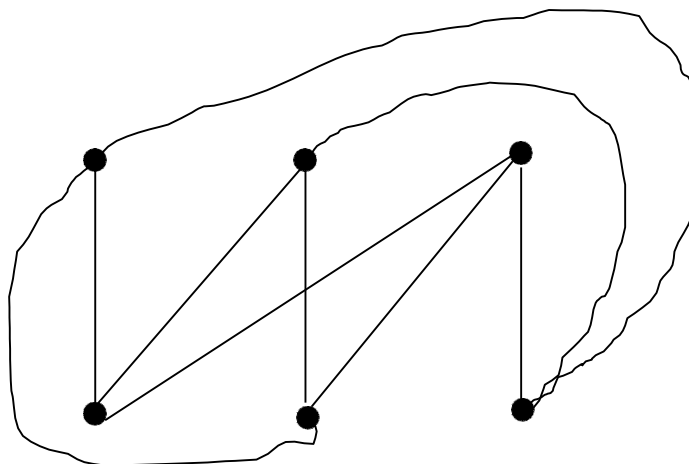
When we have to walk through all the edges and come back to the original vertex i.e., we find an Eulerian circuit in a graph. However, in this graph, the degree of each vertex is not even ($\deg A = \deg B = \deg Q = 3$, $\deg P = 5$) meaning that the Euler circuit does not exist in the graph. G is not an Euler graph. Hence there is no solution to this problem.

2. Utilities problem: There are three houses H_1, H_2 , and H_3 each to be connected to each of utilities-water(W), Gas(G) and electricity(E) by means of conduits (Ex: Channel, pipe or tube)



Is it possible to make such connections without any crossovers of the conduits?

This problem can be represented by a graph as shown below.



The houses and utility supply centers are vertices and the conduits are shown as edges. This graph cannot be drawn in the plane without edges crossing over. Thus the answer to the problem is no.

3. Seating problem: Nine members of a new club meet each day for lunch at a round table. They decide to sit such that every member has different neighbours at each lunch. How many days can this arrangement last?

This situation can be represented by a graph with nine vertices such that each vertex represents a member and an edge joining two vertices represents the relationship of sitting next to each other.

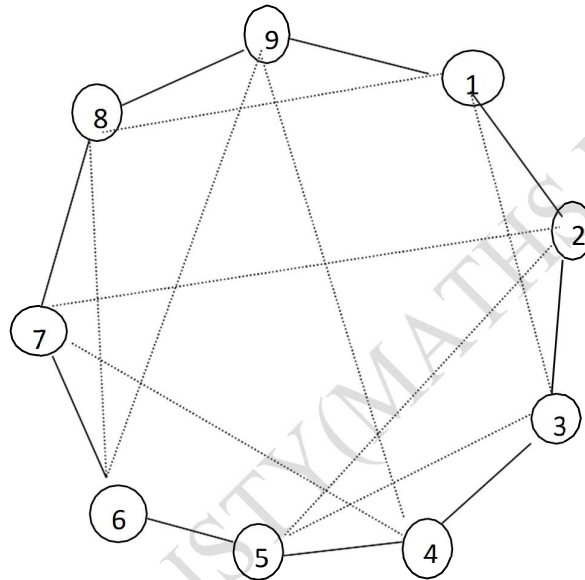


Figure shows there are two possible seating arrangements -1234567891 and 1352749681.

There can be two more arrangements .They are-1573928461 and 1795836241.

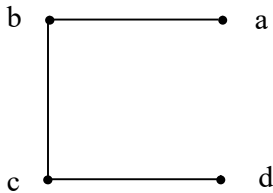
In general it can be shown that for n people, the number of such possible arrangements is $\frac{n-1}{2}$ if n is odd and $\frac{n-2}{2}$ if n is even

Therefore, for the given problem, 4 days of seating arrangement for lunch can make every member have different neighbours each time.

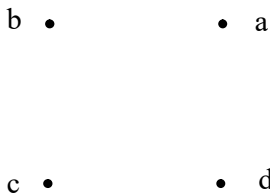
Graph:

A Graph is an ordered pair (V, E) , where V is a non-empty vertex set and E is the edge set.

A graph is usually denoted by $G \equiv (V, E)$.

Example:**Null Graph:**

A graph with no edges is called null graph.

**Trivial Graph:**

A graph with only one vertex is called trivial graph.

**Finite Graph:**

A graph with a finite number of vertices and a finite number of edges is called a finite graph.

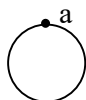
Order and Size of a Graph:

The number of vertices in a graph is called the order, and the number of edges in the graph is called the size of the graph.

A graph of order n and size m is called an (n, m) graph.

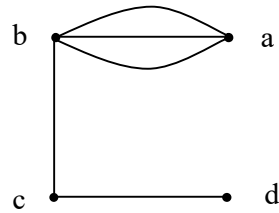
Loop:

A loop is an edge having the same end vertices in a graph.

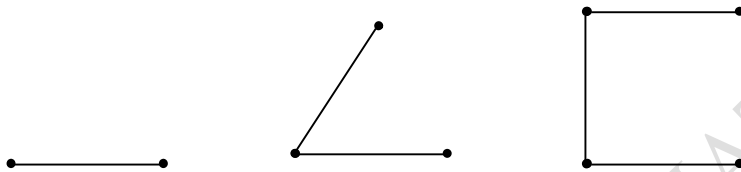


Multiple edges:

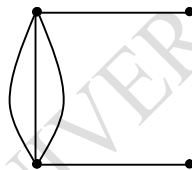
In a graph, if there are two or more edges having the same end vertices are called multiple edges.

**Simple Graph:**

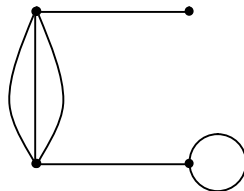
A graph that does not contain any loop and multiple edges is called simple graph.

**Multiple Graph:**

A graph which contains multiple edges but not loops is called multiple graph.

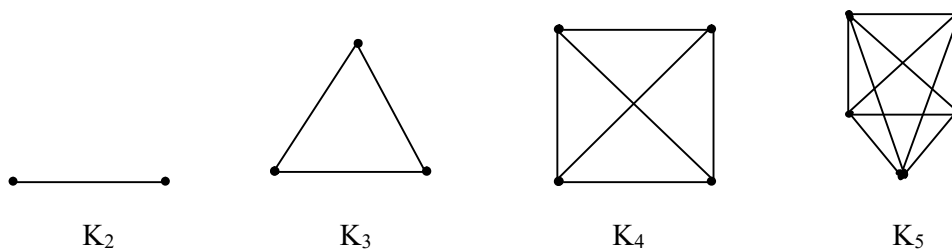
**General Graph:**

A graph which contains both multiple edges and loops is called general graph.

**Complete graph: (Jan⁰⁶ 2009, July⁰⁶ 2009, June 2015)**

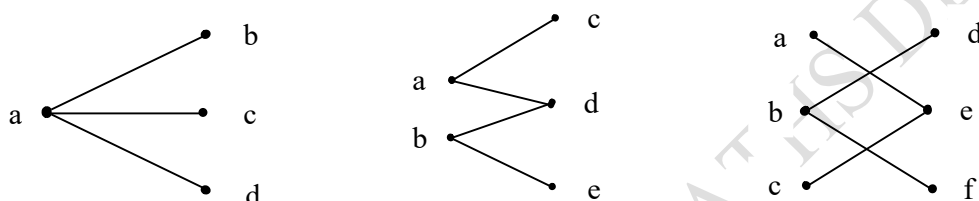
A complete graph is a simple graph with order greater than or equal to two in which there is an edge between every pair of vertices.

A complete graph with order n greater than or equal to two is denoted by K_n . K_5 is Kuratowski's first graph



Bipartite Graph: (July⁰⁶ 2009, Jan⁰⁶ 2010)

A bipartite graph is a simple graph in which the vertex set V is the union of two mutually disjoint non-empty sets V_1 and V_2 which are such that each edge in G joins a vertex of V_1 and the vertex of V_2 .



A bipartite graph is usually denoted by $G = (V_1, V_2, E)$. The sets V_1 and V_2 are called bipartites (or partitions) of the vertex set V .

Note:

In a bipartite graph $V = V_1 \cup V_2$ and $V_1 \cap V_2 = \phi$

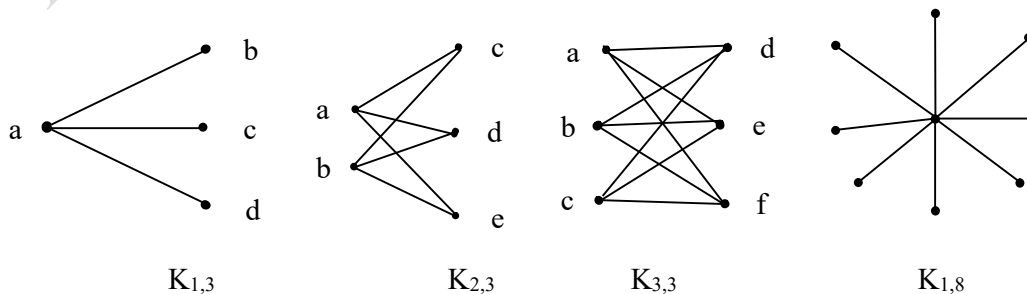
Complete bipartite graphs: (July⁰⁶ 2009, Dec⁰⁶ 2010, Dec⁰⁶ 2011, July 2011)

A complete bipartite graph is a bipartite graph $G = (V_1, V_2, E)$ in which there is an edge between every vertex of V_1 and every vertex of V_2 .

A complete bipartite graph $G = (V_1, V_2, E)$ in which the bipartites V_1 and V_2 contain r and s vertices respectively, with $r \leq s$, is denoted by $K_{r,s}$.

Note: A complete bipartite graph $K_{r,s}$ has $r + s$ vertices and rs edges.

$K_{3,3}$ is called as Kuratowski's second graph.



Example: (July⁰⁶ 2012)

Show that the max number of edges in a simple graph with n vertices is $n(n-1)/2$

Solution:

Each edge of a graph is determined by a pair of vertices. In a simple graph there occur no multiple edges. As such, in a simple graph, the number of edges cannot exceed the number of pair of vertices. The number of pairs of vertices that can be chosen from n vertices is

$${}^nC_2 = \frac{n!}{(n-2)!2!} = \frac{n(n-1)}{2}$$

Thus for a simple graph with n (≥ 2) vertices, the number of edges cannot exceed $\frac{n(n-1)}{2}$

Therefore, if a simple graph has n vertices and m edges, then $m \leq \frac{n(n-1)}{2}$, or $2m \leq n^2 - n$ or

$$2|E| \leq |V|^2 - |V|$$

Example:

Show that a complete graph with n vertices namely K_n has $n(n-1)/2$ edges. Hence find the number of edges in K_{10} .

Solution:

In a complete graph there exists exactly one edge between every pair of vertices hence the number of edges in complete graph is equal to number of pairs of vertices.

If the number of vertices is n , then the number of pairs of vertices is given by ${}^nC_2 = n(n-1)/2! = n(n-1)/2$.

Hence K_n contains $n(n-1)/2$ edges.

Thus K_{10} contains $10(10-1)/2 = 5 \times 9 = 45$ edges.

Example: (Jan 2014)

Show that a simple graph of order $n = 4$ and size $m = 7$ and a complete graph of order $n = 4$ and size $m = 5$ does not exist.

Solution:

For $n = 4$, we have

$$n(n-1)/2 = (4 \times 3)/2 = 6$$

Since $m = 7$ exceeds this number, a simple graph of order $n = 4$ and size $m = 7$ does not exist.

Similarly, since $m = 5$ is not equal to $n(n-1)/2 = (4 \times 3)/2 = 6$, a complete graph of order 4 and size $m = 5$ does not exist.

Example: (July⁰⁶ 2011)

How many vertices and edges are there in the complete graph $K_{4,7}$ and $K_{7,11}$.

Solution:

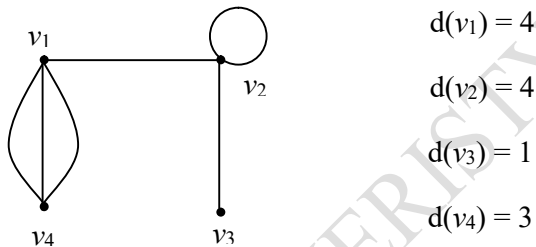
Since $K_{r,s}$ has $r + s$ vertices and rs edges.

Therefore $K_{4,7}$ has $4 + 7 = 11$ vertices and $4 \times 7 = 28$ edges.

$K_{7,11}$ has $7 + 11 = 18$ vertices and $7 \times 11 = 77$ edges.

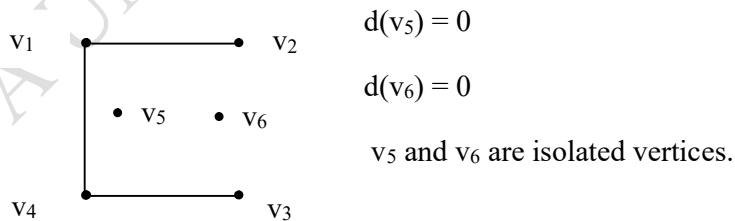
Vertex degree:

In a graph G the number of edges incident on the vertex v with the loops counted twice is called degree of the vertex v .



Isolated Vertex:

A vertex in a graph whose degree is zero is called isolated vertex.

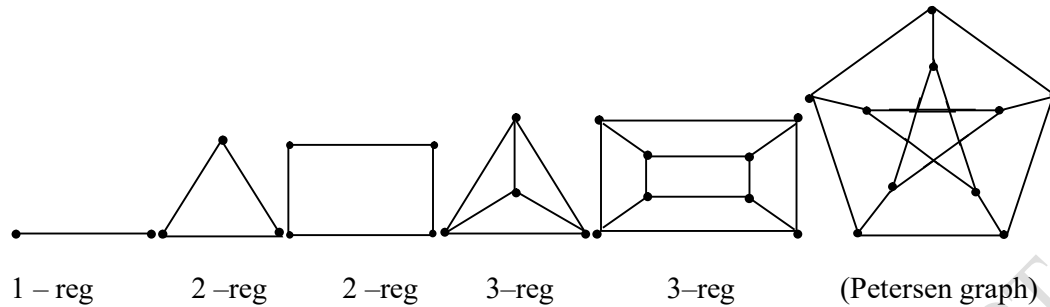


Pendent Vertex:

A vertex whose degree is one is called pendent vertex.

Regular Graph: (Jan⁰⁶ 2009, Dec⁰⁶ 2011, Dec 2012)

A graph in which all vertices are of same degree is called regular graph. If the degree of each vertex is k then it is called k – regular.



Hand shaking property:

The sum of degree of all the vertices in a graph is always even and this number is equal to twice the number of edges in the graph.

Note: If a graph has n vertices and m edges then *i.e.* $\sum_{i=1}^n d(v_i) = 2m$ (even)

Theorem:

In any graph the number of vertices of odd degree is even.

Proof:

Consider a graph G with m edges, n vertices and in which k numbers of vertices are of odd degree. *i.e.* $v_1, v_2, v_3, \dots, v_k$ are of odd degree vertices and the remaining vertices $v_{k+1}, v_{k+2}, v_{k+3}, \dots, v_n$ are of even degree vertices.

$$\text{Then } \sum_{i=1}^n d(v_i) = \sum_{i=1}^k d(v_i) + \sum_{i=k+1}^n d(v_i) \quad (1)$$

In equation (1) the LHS is even by hand shaking property, and the second term in RHS is even because it is the sum even numbers. Hence the first term in RHS must be even.

$$\text{i.e. } d(v_1) + d(v_2) + d(v_3) + \dots + d(v_k) = \text{even} \quad (2)$$

This is possible only when the number of terms in the LHS of the equation (2) is even. Thus in any graph the number of odd degree vertices is even.

Example: (July⁰⁶ 2011)

If G is a graph with n vertices, m edges and k – regular then show that $m = nk/2$.

Solution:

If G is a k – regular graph then every vertex is of degree k . Therefore the sum of the degrees of the vertices is equal to nk , because G contains n vertices and this number is equal to twice the number of edges in the graph.

i.e. $nk = 2m$ (By hand shaking property)

$$\Rightarrow m = nk/2$$

Does there exists a cubic graph of 15 vertices? Does there exists a 4 – regular graph of 15 edges?

(i) Cubic graph means 3 – regular

$$\therefore k = 3, n = 15$$

$$\therefore m = kn/2 = (15 \times 3)/2 = 45/2$$

Since $45/2$ is not a whole number therefore this graph does not exists.

(ii) If G is 4 – 4-regular

$$\therefore k = 4, m = 15$$

$$nk = 2m$$

$$\Rightarrow n \times 4 = 2 \times 15$$

$$\therefore n = 30/4 = 15/2$$

Since $15/2$ is not a whole number therefore this graph does not exists.

Example:

Determine the order of V of the graph $G = (V, E)$ in the following cases

(i) G is a cubic graph with 9 edges.

(ii) G is a regular graph with 15 edges.

(iii) G has 10 edges with 2 vertices of degree 4 and all others are of degree 3.

Solution:

(i) Since G is a cubic graph with 9 edges.

$$\therefore k = 3, m = 9.$$

$$nk = 2m \Rightarrow n = (2 \times 9)/3 = 6$$

$$\therefore |V| = 6.$$

(ii) Since G is regular graph with 15 edges.

Given $m = 15$

$$nk = 2m \Rightarrow k = (2 \times 15)/n$$

$$\Rightarrow k = 30/n$$

Here k is a positive integer and n is positive divisor of 30.

i.e. 1, 2, 3, 5, 6, 10, 15, 30

The possible number of vertices in G is 1, 2, 3, 5, 6, 10, 15, 30.

(iii) G has 10 edges with 2 vertices of degree 4 and all other are of degree 3.

Let G be the graph with n vertices, then by hand shaking property

$$(2 \times 4) + (n - 2)3 = 2 \times 10$$

$$\Rightarrow 8 + (3n - 6) = 20$$

$$\Rightarrow 3n = 18$$

$$\Rightarrow n = 6$$

$$\therefore |V| = 6.$$

Example: (Dec 2014)

Let $G = (V, E)$ be a connected undirected graph. What is the largest possible value for $|V|$ if $|E| = 19$ and $\deg(v) \geq 4$ for all $v \in V$?

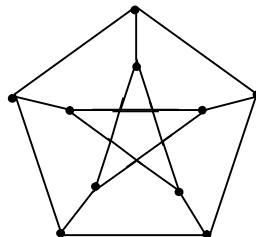
Solution:

Since all vertices are of degree greater than or equal to 4, the sum of the degrees of vertices is greater than or equal to $4n$, where $n = |V|$ is the number of vertices. This sum is equal to twice $|E|$, by handshaking property. Therefore $2|E| \geq 4n$, that is $2 \times 19 \geq 4n$ or $n \leq 38/4 < 10$.

Thus, the largest possible value of $|V|$ is 9.

Example: (Dec⁰⁶ 2011)

Define regular graph. Draw a graph which has 10 – vertices and 15 edges and which should be a 3 – regular graph.



Example: (July⁰⁶ 2012)

How many vertices will the following graphs have if they contain?

(i) 16 edges and all vertices of degree 4.

(ii) 12 edges, 6 vertices of degree 3, and other vertices of degree less than 3.

Solution:

(i) If a graph G is 4-regular, then the degree of every vertex is 4. Therefore, if G has n vertices, then the sum of the degrees of vertices is $4n$. By handshaking property, this sum must be equal to $2m = 2 \times 16 = 32$, since G has 16 edges.

$$\text{Thus } 4n = 32$$

Therefore $n = 8$.

(ii) If a graph G has 12 edges, 6 vertices of degree 3, and other vertices of degree less than 3, then the sum of the degrees of vertices is greater than $(6 \times 3) + (n - 6) \times 3 = 3n$.

By handshaking property, this sum must less than $2m = 2 \times 12 = 24$, since G has 12 edges.

$$\text{Thus } 3n > 24,$$

$$\text{i.e., } n > 8$$

Therefore n is at least 9.

Example: (July⁰⁶ 2009, Dec⁰⁶ 2011)

For a graph with n vertices and m edges, if δ is the minimum and Δ is the maximum of the degrees of vertices then show that $\delta \leq (2m/n) \leq \Delta$.

Solution:

Let $d_1, d_2, d_3, \dots, d_n$ be the degrees of the n vertices of the graph G which contains m edges.

Then $d_1 + d_2 + d_3 + \dots + d_n = 2m$ by hand shaking property.

Since it is given that δ is minimum of $d_1, d_2, d_3, \dots, d_n$

Therefore $\delta = \min$ of $(d_1, d_2, d_3, \dots, d_n)$

$$\Rightarrow d_1 \geq \delta, d_2 \geq \delta, d_3 \geq \delta \dots \dots d_n \geq \delta$$

Therefore $d_1 + d_2 + d_3 + \dots + d_n \geq n\delta$

$$\Rightarrow 2m \geq n\delta \quad (1)$$

And $\Delta = \max (d_1, d_2, d_3, \dots, d_n)$

$$\Rightarrow d_1 \leq \Delta, d_2 \leq \Delta, d_3 \leq \Delta, \dots, d_n \leq \Delta$$

Therefore $d_1 + d_2 + d_3 + \dots + d_n \leq n\Delta$

$$\Rightarrow 2m \leq n\Delta \quad (2)$$

From (1) and (2) we get $n\delta \leq 2m \leq n\Delta$. Dividing by n we get $\delta \leq (2m/n) \leq \Delta$.

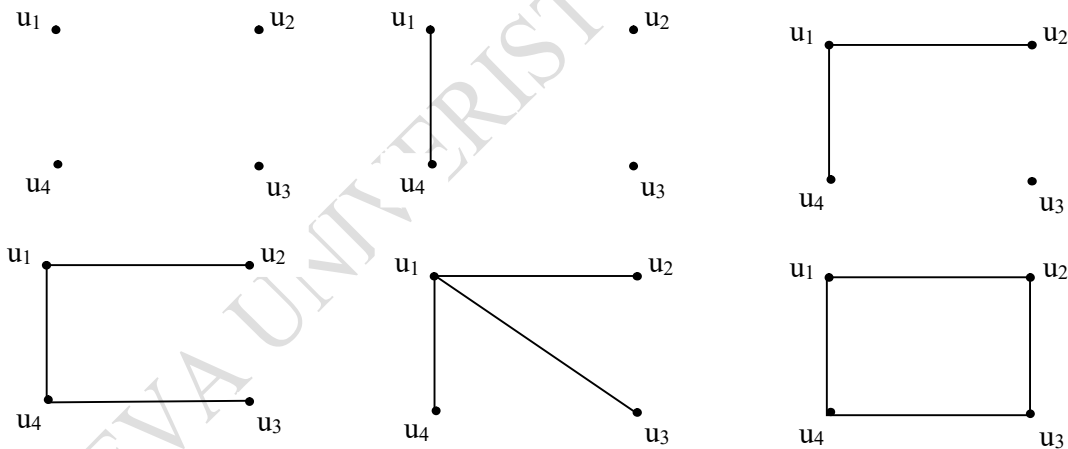
Isomorphism:

Definition: Let $G_1 = \{V_1, E_1\}$ and $G_2 = \{V_2, E_2\}$ be the given graphs, the graph G_1 is said to be isomorphic to G_2 if it satisfies the following conditions

- (i) If there exists a function $f: V_1 \rightarrow V_2$ is both one – one and on to.
- (ii) Whenever there exists an edge AB in G_1 then there must be an edge $\{f(A), f(B)\}$ in G_2

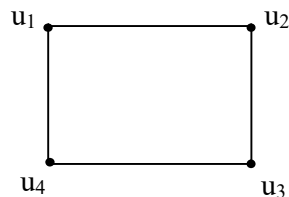
and is denoted by $G_1 \cong G_2$.

Example: Find all (loop free) no-isomorphic undirected graphs with four vertices. How many of these graphs are connected.

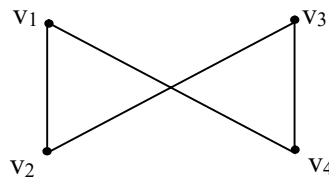


Three graphs are connected.

Example: Prove that the following graphs are isomorphic.



G_1



G_2

Solution:

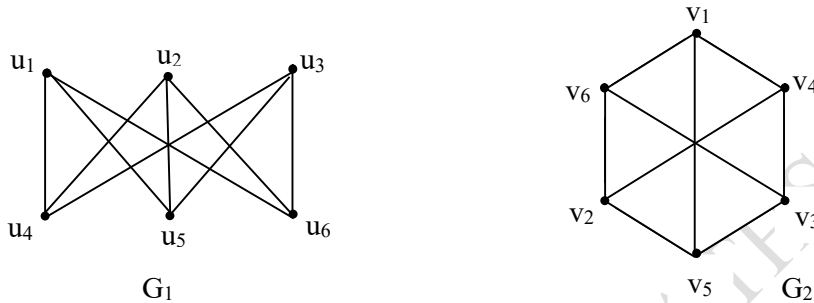
Since there is one to one correspondence between the vertices of the graphs G_1 and G_2 ,

i.e. $u_1 \leftrightarrow v_1, u_2 \leftrightarrow v_2, u_3 \leftrightarrow v_3, u_4 \leftrightarrow v_4$ and

$\{u_1, u_2\} \leftrightarrow \{v_1, v_2\}, \{u_2, u_3\} \leftrightarrow \{v_2, v_3\}, \{u_3, u_4\} \leftrightarrow \{v_3, v_4\}, \{u_1, u_4\} \leftrightarrow \{v_1, v_4\}$

$\therefore G_1 \cong G_2$.

Example: Prove that the following graphs are isomorphic.



Solution: We first note that both the graphs have six vertices each of degree 3, and nine edges. Consider the following 1-1 correspondence between the vertices of these two graphs i.e., $u_i \leftrightarrow v_i$ for $i = 1, 2, 3, 4, 5, 6$.

Under this correspondence the edges in the two graphs correspond with each other as indicated below.

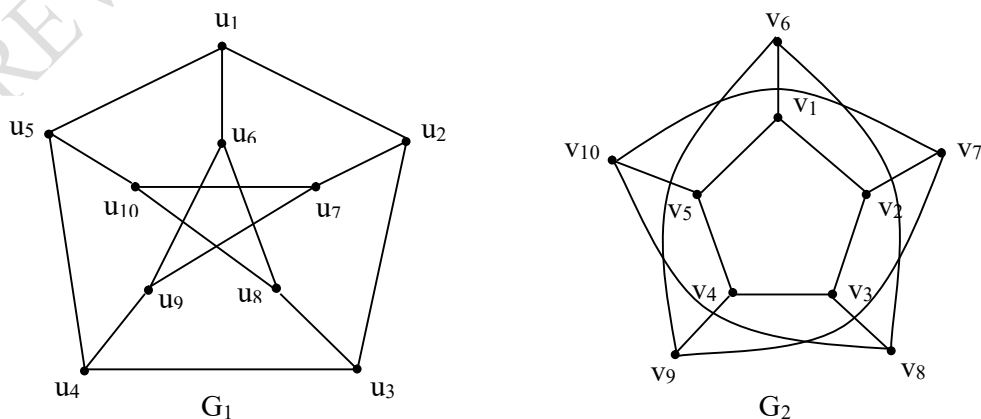
$\{u_1, u_4\} \leftrightarrow \{v_1, v_4\}, \{u_1, u_5\} \leftrightarrow \{v_1, v_5\}, \{u_1, u_6\} \leftrightarrow \{v_1, v_6\},$

$\{u_2, u_4\} \leftrightarrow \{v_2, v_4\}, \{u_2, u_5\} \leftrightarrow \{v_2, v_5\}, \{u_2, u_6\} \leftrightarrow \{v_2, v_6\},$

$\{u_3, u_4\} \leftrightarrow \{v_3, v_4\}, \{u_3, u_5\} \leftrightarrow \{v_3, v_5\}, \{u_3, u_6\} \leftrightarrow \{v_3, v_6\}$

Hence the given graphs are isomorphic to each other.

Example: Show that the following graphs are isomorphic



Solution:

We first note that both the graphs have ten vertices each of degree 3, and 15 edges.

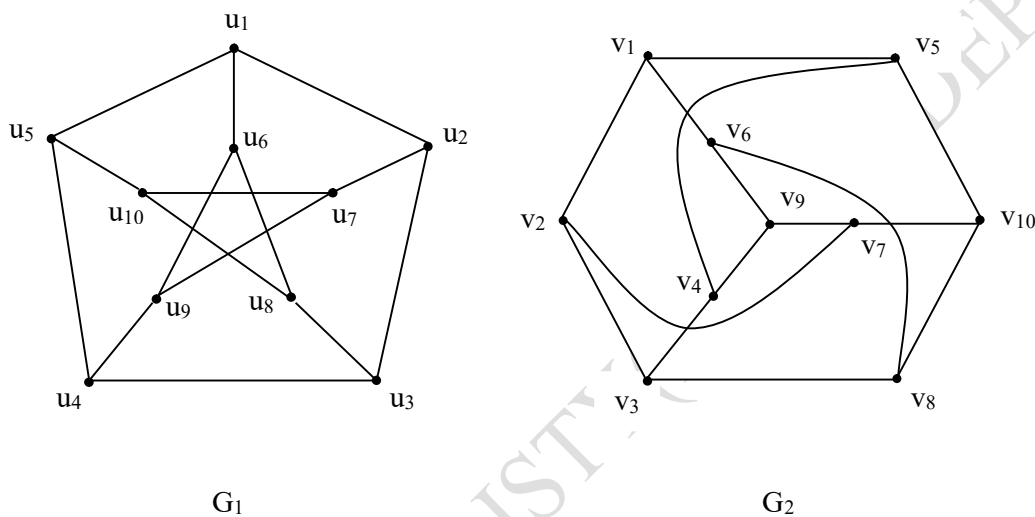
Consider the following 1-1 correspondence between the vertices of these two graphs i.e.,

$u_i \leftrightarrow v_i$ for $i = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$.

There is a one-to-one correspondence between the edges in G_1 and G_2 .

Hence the given graphs are isomorphic to each other ($G_1 \cong G_2$).

Example: Show that the following graphs are isomorphic



Solution:

We first note that both the graphs have 6 vertices each of degree 3, and 9 edges.

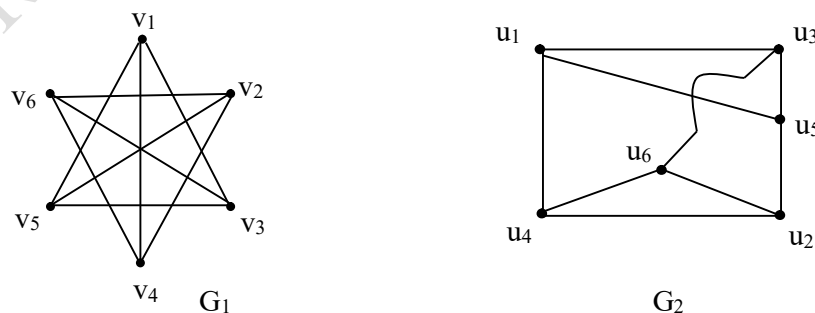
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Example: Verify that the following graphs are isomorphic or not



Solution:

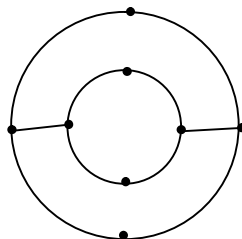
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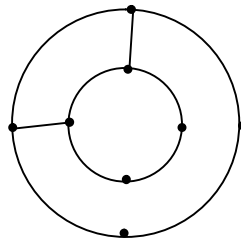
$u_i \leftrightarrow v_i$ for $i = 1, 2, 3, 4, 5, 6$. There is a one-to-one correspondence between the edges in G_1 and G_2 . Hence the given graphs are isomorphic to each other ($G_1 \cong G_2$).

Example: (Dec 2014)

Determine whether the following graphs are isomorphic or not:



G_1



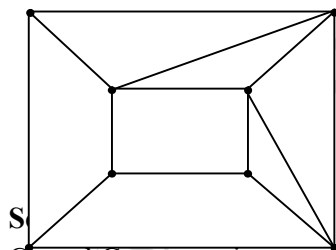
G_2

Solution:

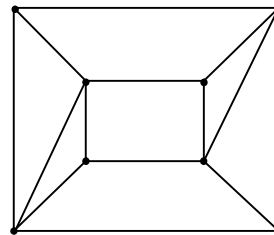
G_1 and G_2 are not isomorphic because G_2 contains 4 pair of vertices of degree 3 are adjacent, where as in G_1 this adjacency is not preserved.

Example:

Verify that the following graphs are isomorphic or not



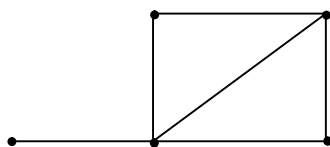
G_1



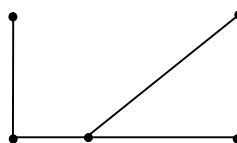
G_1 and G_2 are not isomorphic because G_1 contains a pair of vertices of degree 4 are adjacent, where as in G_2 this adjacency is not preserved.

Example: (June 2008, Jan⁰⁶ 2010)

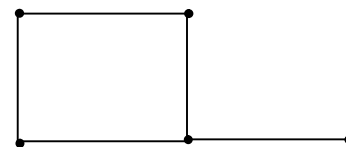
Show that the following graphs are not isomorphic



G_1



G_2



G_3

Solution:

G_1 and G_2 are not isomorphic because these graphs contain unequal edges.

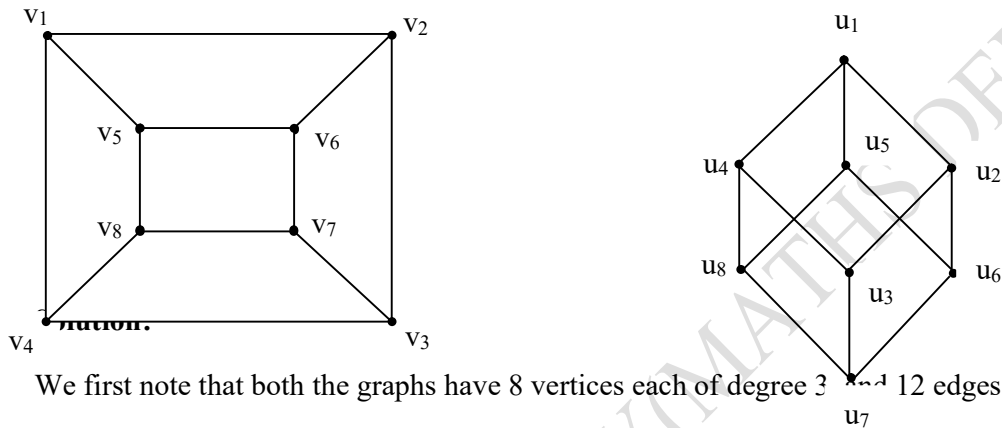
G_1 and G_3 are not isomorphic because these graphs contain unequal edges.

G_2 and G_3 are not isomorphic because degree 3 vertex in G_2 is adjacent to 3 vertices of degree 2 and in G_3 degree 3 vertex is adjacent to 1 degree vertex.

The vertex adjacency is not preserved.

Example:

Verify that the following graphs are isomorphic or not



We first note that both the graphs have 8 vertices each of degree 3 and 12 edges.

Consider the following 1-1 correspondence between the vertices of these two graphs, i.e.,

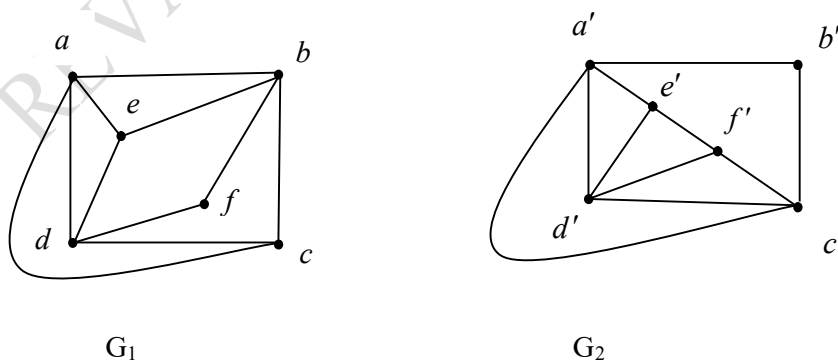
$v_i \leftrightarrow u_i$ for $i = 1, 2, 3, 4, 5, 6, 7, 8$.

There is a one-to-one correspondence between the vertex edges in G_1 and G_2 .

Hence the given graphs are isomorphic to each other ($G_1 \cong G_2$).

Example: (Dec⁰⁶ 2011)

Define isomorphism of graphs. Determine whether the following graphs are isomorphic.

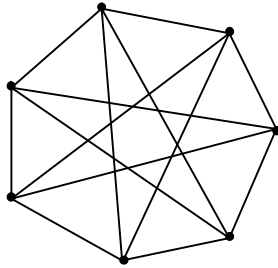


Solution:

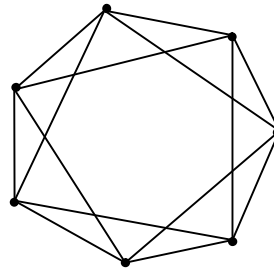
G_1 and G_2 are not isomorphic because G_2 contains one pair of vertices of degree 3 are adjacent, where as in G_1 this adjacency is not preserved.

Example: (Feb 2006)

Show that the following graphs are isomorphic



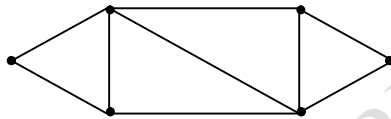
G



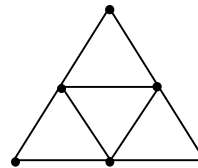
G'

Example:

Verify that the following graphs are isomorphic or not



G_1

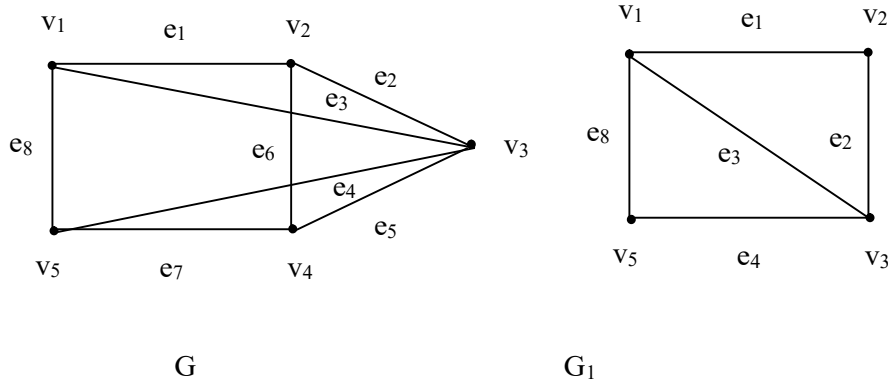


G_2

Subgraph: (June 2015)

A graph $G_1 = (V_1, E_1)$ is said to be the sub graph of $G = (V, E)$ if the following conditions satisfies

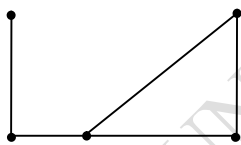
- i) All the vertices and edges of G_1 are in G
- ii) Each edge of G_1 has the same end vertices in G as in G_1 .

Example:**Note:**

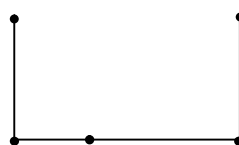
- 1) Every graph is a subgraph of itself.
- 2) Every simple graph of n vertices is a subgraph of the complete graph K_n .
- 3) If G_1 is a subgraph of G_2 and G_2 is a subgraph of G , then G_1 is a subgraph of G .
- 4) A single vertex in a graph G is a subgraph of G .
- 5) A single edge in a graph G , together with its end vertices, is a subgraph of G .

Spanning Subgraph:

A subgraph G_1 of a graph G is said to be Spanning subgraph of G , if G_1 contains all vertices of G .



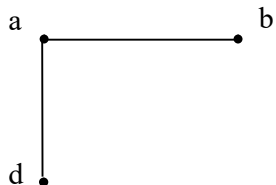
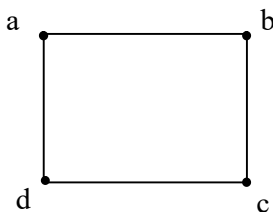
(a)



(b)

Induced Subgraph:

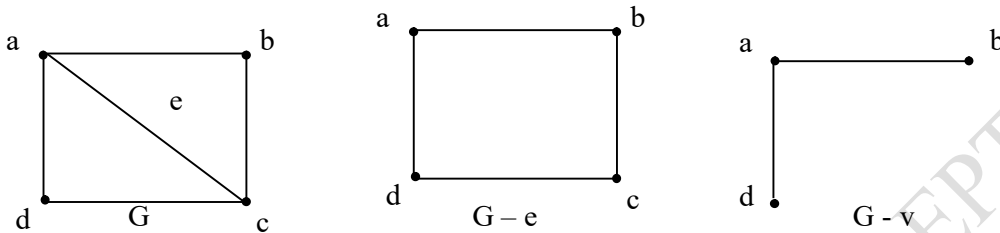
Given a graph $G = (V, E)$, suppose there is a subgraph $G_1 = (V_1, E_1)$ of G such that every edge (A, B) of G , where $A, B \in V_1$ is an edge of G_1 also. Then G_1 is called an induced subgraph of G and is denoted by $\langle V_1 \rangle$.



Edge Disjoint and Vertex disjoint sub graphs:

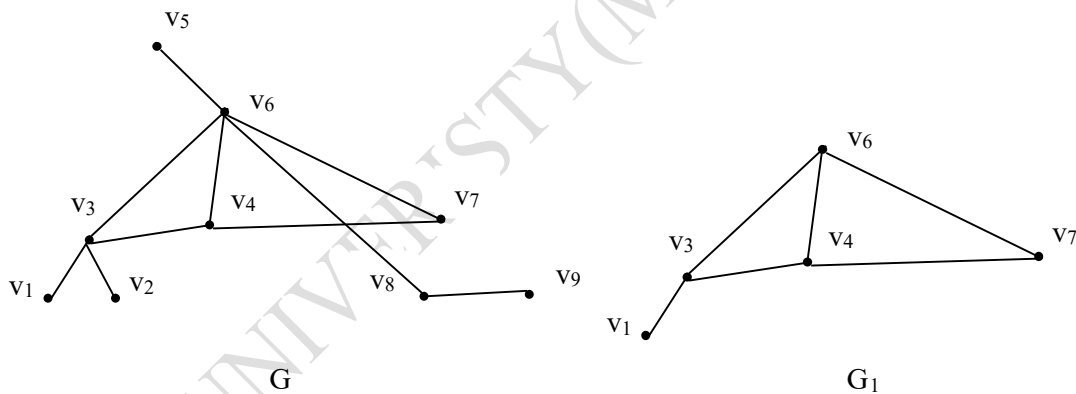
A subgraph $(G - e)$ of a graph G is called the edge disjoint subgraph of the graph G , which is obtained after removing an edge e from the graph G .

A subgraph $(G - v)$ of a graph G is called vertex disjoint subgraph of the graph G , which is obtained by removing a vertex v and the corresponding edges which are incident on vertex.

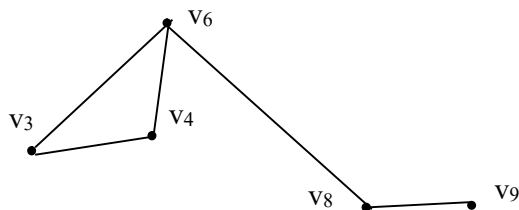
Example:**Example:**

Consider the graph G shown in the figure

- Verify that the graph G_1 shown in the figure is induced subgraph of G . Is this a spanning subgraph of G ?
- Draw a subgraph G_2 of G induced by the set $V_2 = \{v_3, v_4, v_6, v_8, v_9\}$.

**Solution:**

- G_1 is an induced subgraph of G but not spanning subgraph of G .
- $V_2 = \{v_3, v_4, v_6, v_8, v_9\}$



Example: (June⁰⁶ 2010)

Let $G = (V, E)$ be the undirected graph, where $|V| \geq 2$. If every induced subgraph of G is connected, can we identify the graph G ?

Walk and their classification:**Walk:**

A walk of a graph G is a sequence containing alternately vertices and edges. In a walk the vertices and edges can be repeated any number of times.

Open Walk:

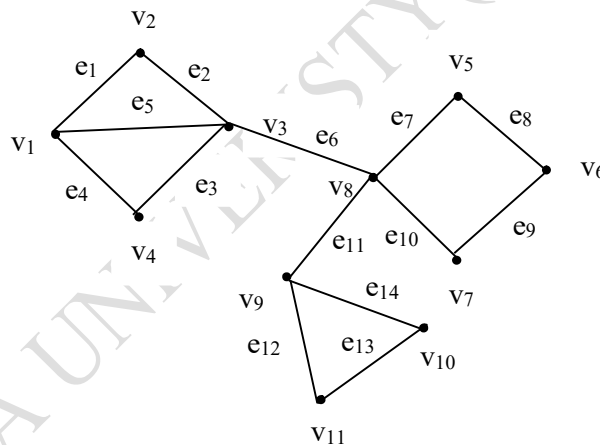
In a walk if the end vertices are distinct then it is called open walk.

Closed Walk:

A walk in which the starting and ending vertex are same is called a closed walk.

Length of the Walk:

The number of edges in a walk is called the length of the walk.

Example:

$v_1e_1v_2e_2v_3e_5v_1e_4v_4e_3v_3e_5v_1$ is a closed walk of length 6.

$v_1e_1v_2e_2v_3e_3v_4e_4v_1e_5v_3e_6v_8e_7v_5e_8v_6e_9v_7e_{10}v_8e_{11}v_9e_{14}v_{10}e_{13}v_{11}e_{12}v_9e_{11}v_8e_6v_3$ is an open walk of length 16

A Trail:

A Trail is an open walk in which no edge is repeated.

Example:

$v_1e_1v_2e_2v_3e_3v_4e_4v_1e_5v_3e_6v_8e_7v_5e_8v_6e_9v_7e_{10}v_8e_{11}v_9$ is a trail of length 11.

Circuit:

A Circuit is a closed walk in which no edge is repeated.

Example: $v_1e_1v_2e_2v_3e_3v_4e_4v_1$ is a circuit of length 4

Path:

A path is an open walk in which no vertex is repeated.

Example: $v_1e_5v_3e_6v_8e_7v_5e_8v_6e_9v_7$ is Path of length 5

Cycle:

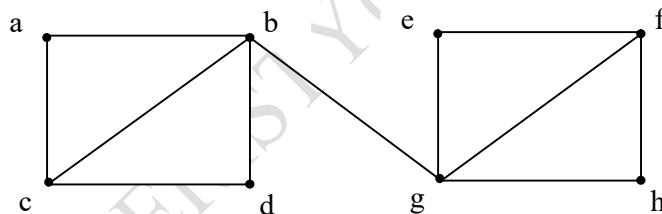
A Cycle is a closed walk in which no vertex is repeated except the starting and ending vertices.

Example: $v_8e_7v_5e_8v_6e_9v_7e_{10}v_8$ is a cycle of length 4.

Example: (June⁰⁶ 2010, June 2012)

Let $G = (V, E)$ be a graph shown in the figure. How many paths are there in G from a – h ?

How many of these paths have a length of 5.

**Solution:**

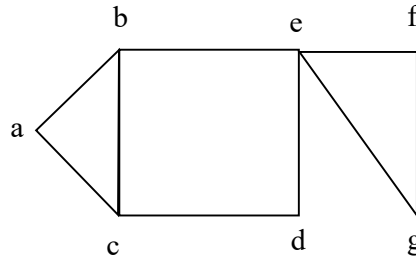
- i) $a \rightarrow c \rightarrow b \rightarrow g \rightarrow h$ of length 4
- ii) $a \rightarrow b \rightarrow g \rightarrow f \rightarrow h$ of length 4
- iii) $a \rightarrow c \rightarrow d \rightarrow b \rightarrow g \rightarrow f \rightarrow h$ of length 6
- iv) $a \rightarrow b \rightarrow g \rightarrow h$ of length 3
- v) $a \rightarrow c \rightarrow d \rightarrow b \rightarrow g \rightarrow h$ of length 5
- vi) $a \rightarrow c \rightarrow b \rightarrow g \rightarrow f \rightarrow h$ of length 5
- vii) $a \rightarrow c \rightarrow d \rightarrow b \rightarrow g \rightarrow e \rightarrow f \rightarrow h$ of length 7
- viii) $a \rightarrow b \rightarrow g \rightarrow e \rightarrow f \rightarrow h$ of length 5
- ix) $a \rightarrow c \rightarrow b \rightarrow g \rightarrow e \rightarrow f \rightarrow h$ of length 6

There are 9 paths from a to h and three paths of these are of length 5.

Example: (June 2015)

For the following graph determine,

- i) A walk from b to d that is not a trail
- ii) A b – d trail that is not a path
- iii) A path from b to d
- iv) A closed walk from b to b that is not a circuit.
- v) A circuit from b to b that is not a cycle
- vi) A cycle from b to b.

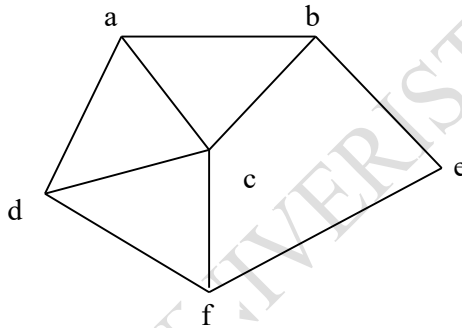


Solution:

- i) befed ii) befged iii) bacd iv) bacdcb v) bacb

Example: (July 2013)

How many different paths of length 2 are there in the undirected graph G in the figure?



Solution:

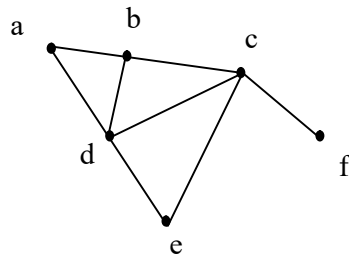
The number of paths of length 2 that pass through the vertex a is the number of pairs of edges incident on a. Since 3 edges are incident on a, thus number is $= {}^3C_2 = 3$.

Similarly the number of paths of length 2 that pass through the vertices b, c, d, e and f are respectively ${}^3C_2 = 3$, ${}^4C_2 = 6$, ${}^3C_2 = 3$, ${}^2C_2 = 1$ and ${}^3C_2 = 3$.

Therefore the total number of paths of length 2 in the given graph is $3+3+6+3+1+3 = 19$

Example: (July⁰⁶ 2008)

In the undirected graph, find (i) an a-a circuit of length 6 (ii) an a-a cycle of maximum length.



(i) An a-a circuit of length 6 is abdcda

(ii) An a-a cycle of maximum length is abceda.

Connected and Disconnected Graphs:

Connected Graph:

A Graph G is said to be connected if there is at least one path between every two distinct vertices in G .

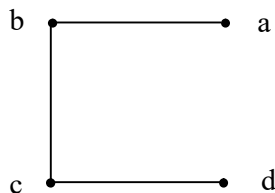
Disconnected Graph:

A Graph G is said to be disconnected if G has at least one pair of vertices between which there is no path.

Example: (Jan⁰⁶ 2009, July 2013, July 2014)

Define connected graph. Give an example of connected graph G where removing any edge of G results in a disconnected graph.

Solution:



Theorem: (Dec 2012)

A connected graph with n vertices has at least $n - 1$ edges.

Proof:

Since the graph is connected $n \geq 2$. If m denotes the number of edges, we have to prove that $m \geq n - 1$, for every positive integer $n \geq 2$. We employ the method of induction to prove this result.

Suppose $n = 2$, then there are exactly two vertices in the graph and since the graph is connected, there must be at least one edge joining these vertices.

Thus now, $m \geq 1 = (2 - 1) = (n - 1)$.

This verifies the required result for $n = 2$.

Assume that the result $m \geq n - 1$ holds for all connected graphs with $n = k$ number of vertices, where k is a positive integer ≥ 2 .

Now, consider a connected graph, say G_{k+1} , with $k + 1$ vertices. Choose a vertex v of this graph and consider the graph G_k obtained by deleting an edge from G_{k+1} for which v is an end vertex. Then G_k is a connected graph with k vertices. Let m_k be the number of edges in G_k . Then from the assumption made in the preceding paragraph, we have $m_k \geq k - 1$.

Consequently $m_k + 1 \geq (k + 1) - 1$.

But $m_k + 1$ is the number of edges in G_{k+1} and $k + 1$ is the number of vertices in G_{k+1} .

Thus, the result $m \geq n - 1$ holds for $n = k + 1$. When it hold for $n = k \geq 2$. Hence by induction the result hold for all integers $n \geq 2$. This completes the proof.

Theorem:

Statement: If a graph has exactly two vertices of odd degree then there must be a path connecting these two vertices.

Proof:

Let v_1 and v_2 be the odd degree vertices in the graph G . Suppose there is no path containing these two vertices then the graph is disconnected and v_1 and v_2 belongs to different components H_1 and H_2 . Consequently each of H_1 and H_2 contain only one vertex of odd degree. This is not possible because H_1 and H_2 are graphs and in a graph the number of vertices of odd degree is always even. Hence there must be a path connecting v_1 and v_2 .

Prove the following:

- (i) A path with n vertices is of length $n-1$.
- (ii) If a cycle has n vertices, it has n edges.
- (iii) The degree of every vertex in a cycle is two.

Solution:

- (i) In a path, every vertex, except the last vertex is followed by precisely one edge. Therefore, if a path has n vertices, it must have $n-1$ edges which is length $n-1$. Therefore, the length is $n-1$.
- (ii) In a cycle, every vertex is followed by precisely one edge. Therefore if a cycle has n vertices, it must have n edges.
- (iii) In a cycle, exactly 2 edges are incident on every vertex(one edge through which we enter the vertex and one edge through which we leave the vertex). Therefore, The degree of every vertex in a cycle is two.